MILITARY SPECIFICATION SHEET

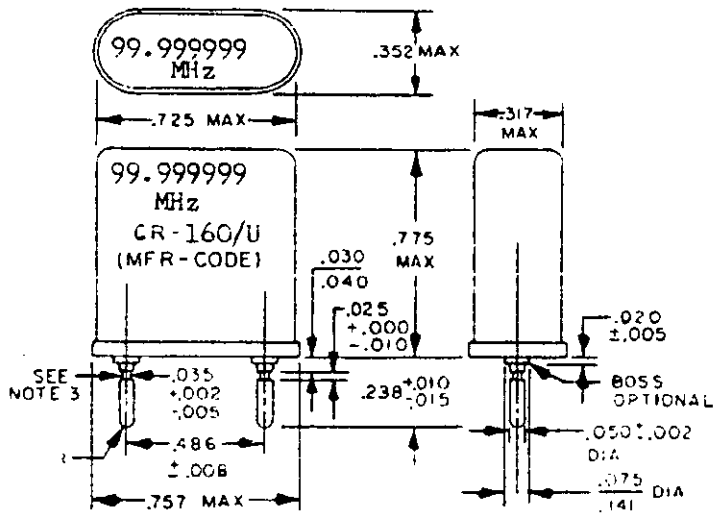
CRYSTAL UNITS, QUARTZ, CR-160/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

(A) INACTIVE FOR NEW DESIGN

Pertinent characteristics: 45 to 100 MHz; fifth overtone; controlled; series.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR-160/U

(A) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
Holder	HC-36-2/U (See MIL-H-10056/23-2)
Dimensions, marking, and configuration	See figure 1
Frequency range	45 to 100 MHz, inclusive
Frequency tolerance	±5 ppm
Equivalent resistance	40 ohms, maximum
Mode of oscillation	Fifth overtone
Temperature ranges:	
Operating (controlled)	70°C to 80°C, inclusive
Operable	-55°C to + 70°C, inclusive
Government reference standard test set	Bridge (See section 6 of MIL-C-3098)
Rated drive level:	1.0 ± 0.2 mW
Equivalent resistance	40 ohms
Resistor voltage drop	0.20 volt
Capacitance, shunt	3 pF maximum
Shock (specified pulse):	
Frequency change permitted	±2.5 ppm
Equivalent resistance	Shall not exceed 40 ohms
Thermal shock:	
Frequency change permitted	±2.5 ppm
Equivalent resistance	Shall not exceed 40 ohms
Vibration:	
Frequency change permitted	±2.5 ppm
Equivalent resistance	Shall not exceed 40 ohms
Aging:	
Frequency change permitted	0.5 ppm

NOTE: CR-160/U is similar to CR-150/U except for holder and tighter aging tolerance.

Ⓐ QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

Ⓐ Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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